

Scalability & Future Plan

Date	02 JULY 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application runs only on a local Flask server.	Cannot be accessed by multiple users over the internet.	High
2	The dataset is static and does not update automatically.	Predictions are limited to the available historical data.	Medium
3	The application does not include user authentication or a database.	User information cannot be securely stored or managed.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports only local users.	Deploy the application on cloud platforms such as Render, Railway, or AWS to support multiple concurrent users.
2	Data Storage	Uses CSV files for input data.	Integrate a relational database such as MySQL or PostgreSQL for secure and scalable data storage.
3	Performance	Predictions are made using a locally stored model.	Optimize the application, implement caching, and use a production WSGI server like Gunicorn or Waitress for better performance.
4	Security	No authentication or user management.	Implement user login, password encryption, and HTTPS for secure access.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Deploy the application to a cloud platform and make it publicly accessible.	1–2 Months	Enables access for multiple users from any location.
Phase 2	Integrate a database and add user authentication with login functionality.	2-3 Months	Improves data management and application security.
Phase 3	Enhance the prediction system with additional financial features, real-time data integration, and improved Machine Learning models.	3-6 Months	Improves prediction accuracy, scalability, and overall user experience.